

Aoxiang Fan

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• aoxiangfan.github.io • PhD candidate in Computer Vision at EPFL

Research Interests

I am dedicated to the study of **3D scene reconstruction and understanding** (related fields: *NeRF*, *Gaussian Splatting*, *Multi-view Stereo*, etc). The goal of my research is primarily to enable machines to better comprehend the world by modeling the environment intrinsically in 3D.

I am also interested and experienced in the matching problem ubiquitously existing in both vision and graphics (related fields: *image matching*, *graph matching*, *point cloud registration*, *shape matching*, etc).

Education

École Polytechnique Fédérale de Lausanne (EPFL), Switzerland	2022-present
Ph.D. in Computer Science, advised by Prof. Pascal Fua	
Electronic Information School, Wuhan University (WHU), China	2018-2021
M.Eng. in Information and Communication Engineering, advised by Prof. Jiayi Ma	GPA:3.91/4.00
Master Thesis (in Chinese): A Study of Robust Algorithms in Image Matching and Its Applications	
Electronic Information School, Wuhan University (WHU), China	2014-2018
B.Eng. in Electronic Information Science and Technology	GPA:3.50/4.00

Selected Publications

- Generalizable Neural Reconstruction of High-Fidelity Surfaces via Sparse Volumetric Representations**
[Aoxiang Fan](#), Corentin Dumery, Nicolas Talabot, Ming Xu, Hieu Le, Pascal Fua
Proc. European Conference on Computer Vision (ECCV), 2026
- A View-consistent Sampling Method for Regularized Training of Neural Radiance Fields**
[Aoxiang Fan](#), Corentin Dumery, Nicolas Talabot, Pascal Fua
Proc. International Conference on Computer Vision (ICCV), 2025
- Counting Stacked Objects**
Corentin Dumery, Noa Ette, [Aoxiang Fan](#), Ren Li, Jingyi Xu, Hieu Le, Pascal Fua
Proc. International Conference on Computer Vision (ICCV), 2025
- Coherent Point Drift Revisited for Non-Rigid Shape Matching and Registration**
[Aoxiang Fan](#), Jiayi Ma, Xin Tian, Xiaoguang Mei, Wei Liu
Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2022
- Image matching from handcrafted to deep features: A survey**
Jiayi Ma, Xingyu Jiang, [Aoxiang Fan](#), Junjun Jiang, Junchi Yan
International Journal of Computer Vision (IJCV), 2021
- Efficient Deterministic Search with Robust Loss Functions for Geometric Model Fitting**
[Aoxiang Fan](#), Jiayi Ma, Xingyu Jiang, Haibin Ling
IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 2021
- Smoothness-Driven Consensus Based on Compact Representation for Robust Feature Matching**
[Aoxiang Fan](#), Xingyu Jiang, Yong Ma, Xiaoguang Mei, Jiayi Ma
IEEE Transactions on Neural Networks and Learning Systems (TNNLS), 2021
- Geometric Estimation via Robust Subspace Recovery**
[Aoxiang Fan](#), Xingyu Jiang, Yang Wang, Junjun Jiang, Jiayi Ma
Proc. European Conference on Computer Vision (ECCV), 2020

Research Projects

A Foundation Model for Fine Geometry Recovery

Supervised by Prof. Pascal Fua

Perspective

○ *Background:* 3D reconstruction has been revolutionized by the advent of foundation models such as VGGT. However, fine-scale reconstruction at a single stage remains a challenge, due to both network architecture and hardware constraints.

○ *Expected Contribution:* We aim to develop a novel method that performs second-stage refinement given rough geometry from foundation models. The method will focus on high-resolution and fine-scale feature learning, employing a sparse volume processing architecture to reduce memory restrictions.

Research Internship

TuSimple-Autonomous Trucking Technology, Beijing

Supervised by Dr. Ji Zhao and Dr. Naiyan Wang

November 2020-March 2021

○ Improved the localization accuracy of the autonomous vehicle by developing an outlier-resilient method for landmark-based 2D-image to 3D-point-cloud alignment.

Technical Strengths

- **Mathematics:** 3D Geometry, Linear Algebra, Numerical Optimization
- **Programming:** Python, C/C++, MATLAB, $\text{\LaTeX}$
- **Developments:** Linux, Git, CMake, Docker, Kubernetes
- **Libraries:** PyTorch, OpenCV, Trimesh
- **Machine Learning:** Neural Rendering, 3D Gaussian Splatting, Generative AI
- **Language Skills:** Chinese (*Native*), English (*Proficient*, TOEFL 108/120), French (*Elementary*)

Awards

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| ○ Recipient of National Encouragement Scholarship of China | 2017 |
| ○ Second Prize Winner of the 17th China Post-Graduate Mathematical Contest in Modeling | 2020 |
| ○ Recipient of Outstanding Master's Thesis by the China Institute of Communications | 2022 |
| ○ Recipient of EDIC Fellowship, EPFL | 2022 |

Additional Experiences

Conference Reviewer

Five-year experience reviewing for : CVPR, ICCV, ECCV, NeurIPS, ICLR.

Teaching Assistant

- CS-442 Computer Vision (*Head TA*) - Lead 8-10 assistants to create course material, teach and evaluate over 190 students. (2023, 2024)
- CS-401 Applied Data Analysis - Help supervise student projects, create course materials and grade exams. The ADA course is one of the largest at EPFL with over 700 students. (2023, 2024, 2025)

Student Supervision

- Boris Zhestiankin (M. Sc.) - *Learning Unsigned Distance Functions from Gaussian Splatting*; Now intern at Inria, Grenoble France.
- Yann Savioz (M. Sc.), - *Gaussian Splatting Applied at City-scale with Lausanne Dataset*; TBD.